but we take nearest y...
it is usefull.

→ Theta Notation

The function $f(n) = \theta(g(n))$ iff ∃ +ve constant c_1, c_2 and n_0

such that $c_1 * g(n) \leq f(n) \leq c_2 * g(n)$

ex. $f(n) = 2n+3$

$$\underbrace{1}_{c_1} * \underbrace{n}_{g(n)} \leq \underbrace{2n+3}_{f(n)} \leq \underbrace{5}_{c_2} \underbrace{n}_{g(n)}$$

$f(n) = \theta(n)$

we cant use any other value

and we should use theta notation

episode - 12 Part 2

ex $f(n) = n^2 \log^n + n$

$1 * n^2 \log^n \leq n^2 \log^n + n \leq 10 n^2 \log^n$

$O(n^2 \log^n)$ as well as $\Omega(n^2 \log^n)$ as well as $\theta(n^2 \log^n)$

ex $f(n) = n!$

$= n \times (n-1)(n-2)(n-3) \cdots 2 \times 1$

In reverse

$= 1 \times 2 \times 3 \cdots (n-1)(n)$

$1 \times 1 \times 1 \cdots 1 \leq f(n) \leq n \times n \times n \times \cdots n$

$1 \leq n! \leq n^n$

$= O(n^n)$

$= \Omega(1)$

we cant take θ as we cant found average bound

ex $f(n) = \log n!$

$\log(1 \times 1 \times 1 \times \cdots) \leq \log(1 \times 2 \times 3 \cdots n) \leq \log(n \times n \times n \times n \times \cdots)$

$1 \leq \log n! \leq \log n^n$

$\leq n \log^n$

$= O(n \log^n)$

$= \Omega(1)$

= No θ for this

if you have to buy a phone then some say you find phone in 20K price may vary 20k - 25k - 18k so this is θ notation

if we give O() figure then upper bond may be 1 lakh, 2 lakh, 5 cores which is not usefull information

if we give Ω figure then lower bond may be 1k, 500 Rs which is also not usefull information

so where ever possible provide theta value as it is giving nearest value and tight bound

but big oh and Ω are used when
you don't have theta

Ep-14 Comparison of functions

Apply log on both sides

ep n^2, n^3 which is greater?

Apply log

$n^2 \quad n^3$

$2 \log^n \quad 3 \log^n$

$2 \log < 3 \log^n$

means

$n^2 < n^3$

this will used in complex functions to find out which is greater

$\log ab = \log a + \log b$

$\log a/b = \log^a - \log^b$

$\log^{a^b} = b \log^a$

$a^{\log_c^b} = b^{\log_c^a}$

$a^b = n$ then $b = \log_a^n$

ex $f(n) = n^2 \log^n$

$g(n) = n(\log^n)^{10}$

$\log[n^2 \log^n] \quad | \quad \log[n(\log^n)^{10}]$

$\log^{n^2} + \log \log n \quad | \quad \log^n + \log(\log^n)^{10}$

$2\log^n + \log\log^n \quad | \quad \log^n + 10 \log \log^n$

So

$f(n) > g(n)$

ex

$f(n) = 3n^{\sqrt{n}}$

$g(n) = 2^{\sqrt{n}\log^n}$

$3n^{\sqrt{n}} \quad | \quad 2^{\sqrt{n}\log_2^n}$

$2^{\log_2 n^{\sqrt{n}}}$

$(n^{\sqrt{n}})^{\log_2^2} \to 1$

$3n^{\sqrt{n}} \quad | \quad n^{\sqrt{n}}$

$f(n) > g(n)$

here without apply log we are able to find out using by formula

ex $f(n) = n^{\log n}$

$g(n) = 2^{\sqrt{n}}$

$n^{\log n} \quad | \quad 2^{\sqrt{n}}$

apply log $\Rightarrow$ $\log^n \log^n \quad | \quad \sqrt{n} \log_2^2$

$(\log n)^2$ | $n^{1/2}$

$\log(\log n)^2$ | $\frac{1}{2}\log n$

apply log again

$2\log\log n$ | $\frac{1}{2}\log n$

this is smaller

$<$

Q $2^{\log n}$ | $n^{\sqrt{n}}$

apply

$\log n \cancel{\log_2 2}$ | $\sqrt{n}\log_2 n$

$\log n$ | $\sqrt{n}\log_2 n$

$<$